

Optimal EV Charging Scheduling by Considering the Limited Number of Chargers

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Abstract—Lack of charging facilities is still a significant barrier to the electrification of the logistic system. How to effectively schedule the electric vehicle (EV) charging power to reduce the charging station operating cost when the number of chargers was limited becomes an important issue. To tackle this issue, an optimal charging scheduling method by responding to the time-of-used (TOU) electricity price is proposed. First, the uncontrolled charging model to fully charge EVs as fast as possible is established. Then, an optimal charging scheduling model by considering the limited number of chargers is proposed to both reduce the charging cost and guarantee the charging demand of each EVs. The proposed model is formulated as a bilevel programming (BP) model. The charger index and available charging duration for each EV are determined at the upper level, while the charging power of each EV at each time slot is determined at the lower level. After that, a solving approach is introduced for the proposed BP model. The efficacy and performance of the proposed charging scheduling method are verified by simulation results.

Index Terms—Bilevel programming (BP), charging scheduling, electric vehicle (EV), genetic algorithm.

NOMENCLATURE

m	Index of charger.
i	Index of electric vehicle (EV).
t	Index of current time slot.
M	Number of charger.
N	Number of charging EVs in one optimization duration.
T	Total number of time slot.
VEC_i	Set of charging information of i th EV.
VEC	Set of sorted EVs.
P_{ch}	Matrix of charging power of each charger in every time slot.

$P_{ch(m,t)}$	Charging power of m th charger at time slot t .
P_{ev}	Matrix of charging power of each EV in every time slot.
$P_{ev(i,t)}$	Charging power of i th EV at time slot t .
δ	Matrix of the chargers assignment.
$\delta_{(k,m)}$	Binary coefficients, which is equal to 1 if the i th EV charge at the m th charger, otherwise, zero.
$e_{price}(t)$	Time-of-used (TOU) electricity price at time slot t .
e_{total}	Total charging cost.
C_i	Index of charger selected by i th EV.
L_i	Position in the charging queue of i th EV.
t_i^a	Arrival time of EV i .
t_i^d	Departure time of EV i .
t_i^s	Time of EV i start charging.
t_i^e	Time of EV i end charging.
t_i^{full}	Minimum time required to fully charge of EV.
P_i^{req}	Requested power of EV i .
P_{max}	Maximum power per time slot provided by one charger.

I. INTRODUCTION

INTEREST in EVs has recently grown due to calls for Ecofriendly transportation. EVs, producing zero tailpipe emissions, offer significant potential in improving sustainability and an ecofriendly environment in urban areas [1]. However, the private EV penetration is still obstructed by the lack of convenient public charging infrastructure, insufficient land for home charging in the metropolitan area, less driving ranges, and long charging time [2], [3]. Compared with increasing the ownership of private EV, electrification of the logistics system is easier to achieve. In the USA, 80% of freight is transported less than 250 mil [4], the front-of-the-pack all-electric trucks are designed to travel on predictable regional or last-mile routes that allow them to return to a central depot for charging. However, the lack of charging facility is still a significant barrier for the logistic companies [5].

In recent years, many studies focusing on the electrified logistics system have been implemented. The advantages of using EVs for the distribution of goods and public transportation in the urban area are discussed in [6] and the use of small-sized EVs and traditional diesel vans on city logistics is compared in [7]. The results show that replacing fossil fuel vehicles by EV can not only benefit the environment but also reduce cargo transportation costs. In order to minimize

Manuscript received August 25, 2020; revised October 13, 2020; accepted October 19, 2020. Date of publication October 26, 2020; date of current version August 24, 2021. This work was supported in part by the German Federal Ministry of Education and Research based upon work in the project KonVeEn under Grant 03EK3057A, in part by the “111” Project of China under Grant B17016, and in part by the China Scholarship Council under Grant 201806130201. (Corresponding author: Jiayan Liu.)

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Digital Object Identifier 10.1109/TTE.2020.3033995

the electric truck energy consumption while guaranteeing the requested battery life, intergraded battery energy management strategies are proposed [8], [9]. EV routing problem is also a research hotspot that its aim is to design appropriate goods delivery path for EV to minimize the overall service cost while subjecting to a variety of state-of-the-art constraints common in EV routing problems [10]–[12]. On this basis, the dynamic characteristic influence of EVs participating in logistics transportation on the power grid is also studied [13], and the corresponding charging dispatch strategy to smooth renewable energy and load fluctuations is also proposed [14].

Charging stations play an important role in providing charging services to EVs. However, the idle parking time of EVs is much longer than the required time to fully recharge the batteries [15], [16]. To reduce the charging cost and improve energy efficiency, extensive research studies focusing on charging scheduling schemes have been carried out. At the power grid level, multiaggregator-based EV charging scheduling methods can manage the charging power in an efficient way to smooth the peak load and reduce the impact of large-scale EVs charging on the power grid [17], [18]. Furthermore, the EV charging station integrated renewable energy is a feasible scenario, and corresponding flexible resource optimization scheduling strategies are investigated to reduce the operating cost [16], [19]. TOU electricity price is not the same in the different periods of one day, therefore the price-sensitive EV charging scheduling methods are proposed for the regulated market [20] and for minimizing the total charging cost and the energy cost from the substation [21]. These works were considered from the perspective of the charging station operating, but the EVs' demand constraints were not fully investigated. To balance the EV users' and charging station operates profit, game theory is adopted in charging power scheduling [22], [23]. However, none of the articles mentioned above considered the realistic situation when a charging station has a limited number of chargers.

For logistics operators with EVs for transportation, charging EVs with the limited number of charging facilities by responding to the TOU electricity price can not only reduce the charging costs for the charging station operators, but also reduce the investment costs. In the realistic charging station, the number of chargers is limited, but the EVs with charging demand will stay in the charging station only for a certain period of time. Thus, it is necessary to arrange the chargers reasonably to complete the EV charging before EVs departure. Considering the coordination of limited charging facilities, optimal charging station planning methods are investigated in [24] and [25]. There is some relevant literature that has considered the limited charging facilities and proposed corresponding optimization schemes [26]–[30]. However, in [26]–[28], the limit on the number of chargers has been simplified to the capacity limit of the charging station, which means they did not consider how to assign specific chargers to each individual EVs while adopting power scheduling. Another method shown in [29] is to reduce the charging station dropping rate considering the limitation of chargers, but this approach treats EVs as a whole and ignores the energy demands of each specific EV.

A model that considers the working status of specific taps is established in [30], however, all the EVs charging in this model are charged with the same formulate, and no specific charging power optimization is carried. To the best of our knowledge, there is almost no published literature on how to assign specific chargers to each individual EVs while adopting power scheduling by responding to TOU price to reduce the total charging cost.

Considering the limited number of chargers, an optimal charging power scheduling method based on TOU electricity price is proposed. First, the uncontrolled charging scheduling (UCS) model is designed for fully charging EVs as fast as possible. There is no coordination among charging EVs, and no charging power optimization scheduling is implemented. Then, considering the limited chargers assignment scheme, an EV optimal charging scheduling model to minimize the total charging cost is proposed. The established model is a bilevel programming (BP) model, which can not only guarantee the EVs' charging demand by reasonably assigning the limited chargers, but also reduce the charging cost by optimal scheduling the charging power. The upper level mainly decides the charger index and available charging period of EVs. The lower level solves the EVs charging power within their available charging period by responding to the TOU electricity price. Then, as the upper level is a mixed nonlinear integer program while the lower level is a linear program, a compound solving algorithm is designed to get the detailed optimal EVs charging scheduling solutions. Through performance verification, the proposed algorithm can find the solution within an acceptable time. Finally, the proposed optimal charging scheduling (POCS) method is compared with the UCS method and a commonly used charging power scheduling method. According to the results, the proposed method can provide a detailed and efficient EV charging scheme, which can minimize the charging cost while guaranteeing the EVs charging demand, when considering the limited number of chargers.

II. SYSTEM MODEL

The POCS strategy is for logistics transportation with EVs. Therefore, in order to meet the requirement of logistics transportation, the requested energy, arrival time, and departure time of EVs are need to be known. When an EV arrives at the charging station, it is required to fill up all the requiring power before its departure time. Due to the limited number of charging facilities, when there is a large number of EVs having charging requirements at the same time, the subsequent EV cannot start charging until the previous EVs finish.

As shown in Fig. 1, while EV₄ arrives at the charging station at t_1 , both of charger₁ and charger₂ are utilized and will finish their work at t_2 and t_3 , respectively. Thus, its available charging time is from t_2 to t_4 at charger₁, and the available charging time is from t_3 to t_4 at charger₂. Moreover, the TOU electricity price in each hour is also different. Therefore, how to schedule the EVs charging power to reduce the charging cost and meet their energy requirements is the optimization problem to be solved.

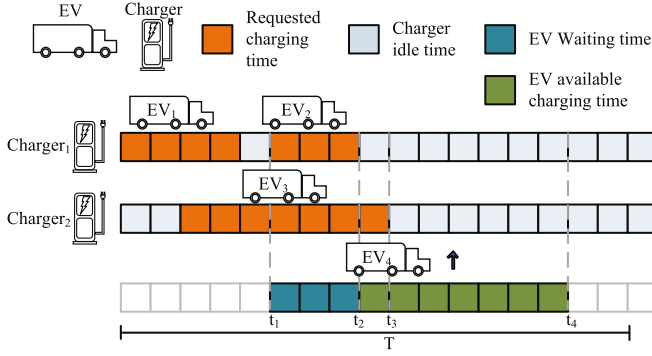


Fig. 1. Charging station operating model.

A. EV Model

Considering the realistic scenarios, the arrival rate and departure rate are the time-dependent data that the EVs distribute goods in the day time and back to the charging station at night. Thus, we set one whole day (24 h, from the first day 12:00 to the next day 12:00) as one period. Then divide it into T time slot equally and let t ($t = 1, 2, \dots, T$) donate t th time slot (an hour is one time slot). The arrival time, departure time, and the required energy of each EVs are required information for optimal scheduling, which is given as follows:

$$VEC_i = \{t_i^a, t_i^d, P_i^{\text{req}}\}. \quad (1)$$

Each arrival EV is sequentially indexed. Considering the realistic logistic transmission scenarios, the number of charging EVs per hour and the required power to each EV are different. The arrival time follows Poisson distribution [29], [31] with an average rate of λ_t . The probability for N EVs arriving at the charging station during time slot t is given by

$$P\{N\} = \frac{e^{-\lambda_t} (\lambda_t)^N}{N!}. \quad (2)$$

Meanwhile, the departure time follows the truncated normal distribution [30], [32] $N(\lambda_1, \mu_1)$, where λ_1 is the mean of the departure time and μ_1 is the standard deviation. The required charging energy is determined by the initial battery state of charge (SOC) S_i^{ini} and the requested SOC S_i^{req} . According to the realistic transportation, the initial SOC and requested SOC are following the normal distribution [32] $N(\lambda_2, \mu_2)$ and $N(\lambda_3, \mu_3)$, respectively, where λ_1 and λ_2 are the mean of the S_i^{ini} and S_i^{req} , and μ_2 and μ_3 are the deviations. Consequently, the required energy for each EV is computed by

$$P_i^{\text{req}} = (S_i^{\text{req}} - S_i^{\text{ini}}) \cdot \text{Cap}. \quad (3)$$

Considering the fact that the vehicle type of a logistics company is uniform, it is supposed that all the types of EVs and chargers are the same and each charger provides the same amount of energy in a certain time slot.

The maximum power that a charger can provide within one time slot is defined as P_{max} . And when an EV charging with P_{max} , the charging time for this EV is the shortest. Thus, the minimum required time to fully charge is

described as

$$t_i^{\text{full}} = \left\lceil \frac{P_i^{\text{req}}}{P_{\text{max}}} \right\rceil \cdot t. \quad (4)$$

Remark 1: Considering the realistic operation of the charging station, it is assumed that when EVs need not queue for charging, all EVs can complete charging within their dwell time at the charging station. Thus, the charging information of i th EV should satisfy: $t_i^d - t_i^a \geq t_i^{\text{full}}$.

Remark 2: The index of EV will be sorted according to the time of arrival. EVs arriving earlier will be charged first. When several EVs have the same arrival time t_i^a , the EV with earlier departure time t_i^d has more urgent charging demand, thus, has higher charging priority. In addition, for EVs with the same arrival time t_i^a and departure time t_i^d , EVs with more energy requirement P_i^{req} will be charged first. The sorted set of EVs according to the above rules is defined as

$$VEC = \{VEC_1, \dots, VEC_i, \dots, VEC_N\}. \quad (5)$$

B. Uncontrolled Charging Station Operation Model

In this article, each single time slot t is an hour. In realistic uncontrolled scenarios, EVs tend to finish charging as fast as possible. Therefore, an EV will immediately start charging with the maximum charging power once a charger was idle.

According to Remark 2, the index order in VEC is the sequence of EVs into the charging station, the charging process is shown in Algorithm 1. Defining the output of Algorithm 1 is the charger power matrix P_{ch} and the EV power matrix P_{ev} , as described in

$$P_{\text{ch}} = \begin{bmatrix} P_{\text{ch}(1,1)} & \cdots & P_{\text{ch}(1,T)} \\ \vdots & \ddots & \vdots \\ P_{\text{ch}(M,1)} & \cdots & P_{\text{ch}(M,T)} \end{bmatrix} \quad (6)$$

$$P_{\text{ev}} = \begin{bmatrix} P_{\text{ev}(1,1)} & \cdots & P_{\text{ev}(1,T)} \\ \vdots & \ddots & \vdots \\ P_{\text{ev}(N,1)} & \cdots & P_{\text{ev}(N,T)} \end{bmatrix}. \quad (7)$$

$P_{\text{ch}} (M \times T)$ is the charger charging power matrix where $P_{\text{ch}(m,t)}$ means the charging power provided by the m th charger at time slot t . Similarly, $P_{\text{ev}} (N \times T)$ is the EV charging power matrix where $P_{\text{ev}(i,t)}$ means the charging power of the i th EV at time slot t .

In the uncontrolled charging model, EVs always select the earliest idle charger to charge. For example, an EV arrives at the charging station while one charger is idle and the other is utilized; this EV will select the idle one. In another case, as shown in Fig. 1, the last EV₄ arrives while all the two chargers are not idle. Because when charger₁ finishes the charging earlier, EV₄ will queue in charger₁.

Remark 3: If the EV i has completed charging at charger m at time slot t , and the charging power provided by the charger m at this time slot is $P_{\text{ch}(m,t)} = P_{\text{ev}(i,t)}$, the maximum allowed charging power for the next EV j who charged at the same charger m at the same time slot t is: $\bar{P}_{\text{ev}(j,t)} = P_{\text{max}} - P_{\text{ev}(i,t)}$.

Remark 4: The under charging EVs will not stop charging until their energy requirements are satisfied or their departure time is reached.

The uncontrolled charging model includes two procedures: idle chargers searching (ICS) procedure and EV power supplement (EPS) procedure. The ICS finds the earliest idle charger λ and its idle time T_{ear} according to the current charging station status P_{ch} . The EPS procedure supplies energy to each EV. For each EV, it will queue at the earliest idle charger λ and start charging after T_{ear} . The output of Algorithm 1 is the EVs charging power matrix P_{ev} and chargers charging power matrix P_{ch} . Thus, the charging characteristics of one charging station under the uncontrolled charging model can be obtained.

Algorithm 1 Uncontrolled Charging Model

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1: Input:  $VEC, M, N, T, P_{max}$ 
2: Output:  $P_{ch}, P_{ev}$ 
3: Initialization:  $P_{ch} = \text{zeros}[M, T]$  ( $M \times T$  null matrix);
 $P_{ev} = \text{zeros}[N, T]$  ( $N \times T$  null matrix)
4: ProcedureEPS( $P_{ev}, P_{ch}, M, T, P_{max}$ )
5: for every EV  $i = 1: N$ 
6:    $(T_{ear}, \lambda) = \text{ICS}(P_{ch}, M, T)$ 
7:   while  $\delta$  is not empty and  $P_i^{req} > 0$  and  $T_{ear} \leq t_i^d$ 
8:     if  $P_{max} - P_{ch}(\lambda, T_{ear}) \leq P_i^{req}$  then
9:        $P_{ch}(\lambda, T_{ear}) = P_{max}$ 
10:       $P_{ev}(i, T_{ear}) = P_{max} - P_{ch}(\lambda, T_{ear}) + P_{ev}(i, T_{ear})$ 
11:       $P_i^{req} = P_i^{req} - (P_{max} - P_{ch}(\lambda, T_{ear}))$ 
12:    else
13:       $P_{ch}(\lambda, T_{ear}) = P_{ch}(\lambda, T_{ear}) + P_i^{req}$ 
14:       $P_{ev}(i, T_{ear}) = P_i^{req} + P_{ev}(i, T_{ear})$ 
15:       $P_i^{req} = 0$ 
16:    end if
17:     $T_{ear} = T_{ear} + 1$ 
18:  end while
19: end for
20: return:  $P_{ch}, P_{ev}$ 
21: end Procedure
22: Procedure ICS( $P_{ch}, M, T$ )
23: Initialize:  $T_{ear} = T$ 
24: for every charger  $m = 1: M$ 
25:   for every time slot  $t = 1: T$ 
26:     if  $P_{ch}(m, t) < P_{max}$  then
27:       When the  $m$ -th charger is idle  $CT_i = t$ 
28:       jump to 31
29:     end if
30:   end for
31:   if  $T_{ear} > CT_i$  then
32:      $T_{ear} = CT_i$ 
33:     Record the earliest idle charger  $\lambda = m$ 
34:   end if
35: end for
36: return:  $T_{ear}, \lambda$ 
37: end Procedure

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III. OPTIMAL CHARGING SCHEDULING

Minimizing the charging cost by optimal charging scheduling can effectively reduce the cost of logistics transportation with EVs. By considering the TOU electricity price, shifting the charging peak into the valley TOU price can reduce the charging cost significantly.

Limited by the number of chargers, the optimal charging process needs to assign chargers to each EV, arrange feasible charging durations for each EV, and determine the charging power of each EV. Therefore, this article decomposes the optimization problem into a BP model. In the upper level, according to EVs' dwell time and request power, each EV is assigned to specific charger, and then its start charging time and end charging time are determined. In the lower level, with the fixed start charging time and end charging time, the charging power of each EV is optimized to minimize the charging cost by responding to the TOU price.

A. Lower Level Model

The lower level model is designed to optimize the charging power based on the TOU electricity price when the charging duration of each EV is determined. Define the i th EV start charging time as t_i^s and the end charging time as t_i^e .

The lower level model is designed as

$$\min_{P_{ev(i,t)}} \sum_{i=1}^N \sum_{t=1}^T P_{ev(i,t)} e_{\text{price}}(t) \quad (8)$$

subject to

$$\sum_{t=t_i^s}^{t_i^e} P_{ev(i,t)} = P_i^{req}, \quad \forall i \in N \quad (9a)$$

$$P_{ev(i,t)} = 0, \quad \forall t \notin [t_i^s, t_i^e], \quad \forall i \in N \quad (9b)$$

$$P_{ev(i,t)} \leq P_{max}, \quad \forall t \in T, \forall i \in N \quad (9c)$$

$$P_{ev(i,t_i^e)} + P_{ev(j,t_i^e)} \leq P_{max}, \quad \text{if } t_i^s = t_j^e. \quad (9d)$$

In the objective function (8), $e_{\text{price}}(t)$ donates the electricity price at time slot t . $P_{ev(i,t)}$ is the decision variable which represents charging power of EV i at time slot t . Constraint (9a) expresses that each EV should be fully charged with the required energy P_i^{req} within its start charging time t_i^s and end charging time t_i^e (Remark 4). Constraint (9b) represents that the i th EV cannot be charged while the time is early than its start charging time t_i^s or later than its end charging time t_i^e . Constraint (9c) means the power provided per time slot cannot exceed the maximum allowed charging power P_{max} . In constraint (9d), in a certain time slot, subsequent EV can be charged with the remaining energy if the charging power of the previous EV at the same time slot has not reached the maximum allowed charging power P_{max} (Remark 3). It can be found that once the charging duration t_i^s and t_i^e are fixed, the lower level model becomes a linear optimization problem.

B. Upper Level Model

Due to the limited number of chargers, the charger assignment process needs to take into account. Define the charger assignment matrix δ ($M \times N$), and its inside elements are described as

$$\delta_{(m,i)} = \begin{cases} 1 & \text{if EV } i \text{ charging in charger } m \\ 0 & \text{otherwise} \end{cases} \quad \forall \delta_{(m,i)} \in \delta. \quad (10)$$

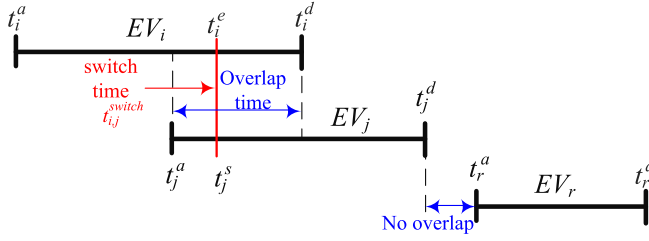


Fig. 2. Example of time limit.

Define C_i as the index of charger selected for i th EV, L_i as the position in the charging queue of i th EV, as described

$$\begin{cases} C_i = m, & \text{if } \delta_{(m,i)} = 1 \\ L_i = \sum_{j=1}^i \delta_{(m,j)}, & \text{if } \delta_{(m,i)} = 1. \end{cases} \quad (11)$$

Therefore, it is necessary to assign each EV to reasonably choose the charger to achieve the purpose of reducing costs. In addition, the start and end charging time should be optimized when the charger assignment is determined.

As shown in Fig. 2, suppose three EVs are assigned to charge at one specific charger, and they have their own arrival and departure times. The first vehicle EV_i is still in charging station when the second vehicle EV_j arrives, thus there is an overlap time between the two EVs. Because one charger can only serve single EV at once, the switch time $t_{i,j}^{switch}$ between EV_i and EV_j should be set. Switch time is the time when the previous EV_i finished charging and the next EV_j started charging, where $t_{i,j}^{switch} = t_i^e = t_j^s$. The end charging time of EV_i is set as t_i^e and this EV can charge during $[t_i^a, t_i^e]$. Similarly, EV_j needs to wait for EV_i to complete charging before starting charging, and the available charging duration of EV_j is $[t_j^s, t_j^d]$. The end charging time of vehicle i is equal to the start charging time of EV_j . For the third arriving EV_r , it has no overlapping dwell time with the previous vehicle, so the available charging duration is $[t_r^a, t_r^d]$.

Selecting the reasonable charger and deciding the suitable start charging and end charging time is in order to reduce the charging cost as much as possible. The mathematical program of the upper level model is

$$\min_{\delta_{(m,i)}, t_i^s, t_i^e} \sum_{i=1}^N \sum_{t=t_i^a}^{t_i^e} \sum_{m=1}^M \delta_{(m,i)} P_{ev(i,t)} e_{price}(t) \quad (12)$$

subject to

$$\sum_{m=1}^M \delta_{(m,i)} \equiv 1, \quad \forall i \in N \quad (13a)$$

$$\begin{cases} t_i^s \geq t_i^a \\ t_i^e \leq t_i^d \end{cases} \quad \forall i \in N \quad (13b)$$

$$t_i^s \leq t_i^e, \quad \forall i \in N \quad (13c)$$

$$\begin{cases} t_j^a \leq t_j^s \leq t_i^e \leq t_i^d, & \text{if } C_i = C_j \text{ \& } L_i + 1 = L_j \text{ \& } t_j^a \leq t_i^d \\ \begin{cases} t_i^s = t_i^a \\ t_i^e = t_i^d \end{cases}, & \text{otherwise} \end{cases} \quad (13d)$$

$$t_i^e - t_i^s \geq t_i^{full}, \quad \forall i \in N. \quad (13e)$$

In order to reduce the total charging cost, objective (12) is set the charger selection element $\delta_{(m,i)}$, start charging time t_i^s , and end charging time t_i^e as the decision variable. Constraint (13a) expresses that all the vehicle should be assigned to one and only one charger for charging. Thus, in each column of matrix δ , at least and only one element is 1. Considering the realistic situation of the charging process, EVs can only be charged while dwelling in the charging station. Therefore, the start charging time and end charging time should satisfy constrain (13b). Constraint (13c) means the end charging time is later than the start charging time. The start charging time of the subsequent j th EV cannot be earlier than the end charging time of the previous i th EV if both i th EV and j th EV charged are at the same charger. Thus, constraint (13d) expresses the time restriction setting when two EVs had overlap dwell time in the charging station. In one case, the start charging time of the subsequent j th EV is equal to the end charging time of the previous i th EV when satisfied: 1) both i th EV and j th EV charge at the same charger ($C_i = C_j$); 2) the j th EV queues just behind the i th EV ($L_i + 1 = L_j$); and 3) the departure time of the previous i th EV is later than the arrival time of the subsequent j th EV ($t_j^a < t_i^d$). Otherwise, the start and end charging times of an EV are set as its arrival and departure time, respectively. Sufficient charging time is the necessary condition for the fulfill charging, as described in constraint (13e).

C. Solving Algorithm

Since the decision variables in the upper-level model are discontinuous integers, and the decision variables t_i^s and t_i^e are even the variable index of the lower-level model, which makes the BP problem very difficult to solve. The definition of t_i^s and t_i^e is to make the charging power in the lower model satisfy the constraint (9a). In fact, it is negligible to first know when an EV starts and finished charging. We only need to ensure that each EV meets the corresponding charging demand within its dwell time and will not replenish energy during nondwell time. Then according to Remark 3, compute the start and end charging time of EVs. Thus, we convert the problem from solving $P_{ev(i,t)}$ into solving $P_{ch(m,t)}$. The converted lower level model is

$$\min_{P_{ch(m,t)}} G = \sum_{m=1}^M \sum_{t=1}^T P_{ch(m,t)} e_{price}(t) \quad (14)$$

subject to

$$\begin{cases} \sum_{t=t_i^a}^{t_i^d} P_{ch(m,t)} \leq P_i^{req} \\ \sum_{t=t_j^a}^{t_j^d} P_{ch(m,t)} \leq P_j^{req} \end{cases} \quad \text{if } \delta_{(m,i)} = \delta_{(m,j)} = 1 \text{ \& } L_i + 1 = L_j, \quad \forall m \in M \quad (15a)$$

$$P_{ch(m,t)} \leq P_{max}, \quad \forall t \in T, \forall m \in M \quad (15b)$$

$$\delta_{(m,i)} P_{ch(m,t)} = 0, \quad \forall t \notin [t_i^a, t_i^d], \forall i \in N, \forall m \in M \quad (15c)$$

$$\sum_{t=t_i^a}^{t_i^d} P_{ch(m,t)} = P_i^{req}, \quad \text{if } \delta_{(m,i)} = 1, \forall i \in N, \forall m \in M. \quad (15d)$$

In the converted lower level, decision variables are transferred from the charging power of EV i at time slot t $P_{ev(i,t)}$ to the charging power of charger m at time slot t $P_{ch(m,t)}$, which satisfy $P_{ch(m,t)} = \sum N i (\delta_{(m,i)} P_{ch(i,t)})$. The realistic meaning is the charging power of one charger is the summation of all the EV power charged in this charger. Constraint (15a) determines the charging duration of EVs with overlapping dwell time. The charging power cannot exceed the required energy of the previous i th EV when the arrival time of the subsequent j th EV t_j^a has not reached. Similarly, the charging power cannot exceed the required energy of the subsequent j th EV when the departure time of the previous i th EV t_i^d already reached. Constraint (15b) is the charging power limited, constraint (15c) represents the fact that EVs cannot be charged when not in the dwell time, and constraint (15d) expresses that EVs should finish charging during their dwell time.

As decision variables in lower level model are converted, the converted upper-level model only needs to consider the chargers assignment matrix δ as decision variables

$$\min_{\delta_{(m,i)}} F = \sum_{i=1}^N \sum_{t=1}^T \sum_{m=1}^M \delta_{(m,i)} P_{ch(m,t)} e_{\text{price}}(t) \quad (16)$$

subject to

$$\sum_{m=1}^M \delta_{(m,i)} \equiv 1, \quad \forall i \in N, \quad \forall m \in M. \quad (17)$$

By the above modification, the decision variables in the upper-level problem are reduced from t_i^s , t_i^e , and $\delta_{(m,i)}$ to only $\delta_{(m,i)}$. Meanwhile, the lower level problem is a linear problem once the decision variables in upper-level problem are fixed. Although this BP problem is still an NP-hard problem, the converted model has greatly reduced the search range compared to the previous model. Currently, heuristic algorithm is one of the mainstream methods for solving such NP-hard problem [27], [30]. Since the decision variables $\delta_{(m,i)}$ are in binary, we do not need to proceed with the encoding and decoding steps when using genetic algorithms. Thus, the genetic algorithm with specific improvements is used to solve this problem. Since the generated matrix δ may conduct to one charger serving for too many EVs, which making the lower level model unsolvable, we need to quickly determine whether the lower-level model can be solved with the generated δ matrix.

Theorem 1: define that the dwell time of K EVs overlaps when charging at the same charger m . Suppose the index of K EVs is $i = 1, 2, \dots, K$. The dwell time overlap means: $(t_1^d < t_2^a) \& \dots \& (t_K^d < t_{K-1}^a)$. Define condition

$$\Pi : \left\{ (t_K^d - t_1^a) P_{\max} < \sum_{i=1}^K P_i^{\text{req}} \mid \exists (t_1^d < t_2^a) \wedge \dots \wedge (t_K^d < t_{K-1}^a) \right\}. \quad (18)$$

Θ : problems (14) and (15) are not solvable.

Π is a necessary and sufficient condition for Θ , which is $\Pi \Leftrightarrow \Theta$. Condition Π can be used to quickly judge whether the lower-level model severable.

Proof 1: First, proof $\Pi \Rightarrow \Theta$ is true. According to condition Π , due to the dwell time of the previous EVs and subsequent EVs overlap, we get

$$\sum_{t=t_1^a}^{t_1^d} P_{ch(m,t)} + \dots + \sum_{t=t_K^a}^{t_K^d} P_{ch(m,t)} = \sum_{t=t_1^a}^{t_K^d} P_{ch(m,t)}. \quad (19)$$

According to constraint (16c) and Π , we have

$$\sum_{t=t_1^a}^{t_K^d} P_{ch(m,t)} \leq (t_K^d - t_1^a) P_{\max} < \sum_{i=1}^K P_i^{\text{req}}. \quad (20)$$

To make the problem solvable, all constraints must be satisfied. But by combining (19) and (20), constraint (15d) cannot be established, which means the problem is not solvable, $\Pi \Rightarrow \Theta$ is true.

Then, proof $\Theta \Rightarrow \Pi$ is true. We can prove $\neg \Pi \Rightarrow \neg \Theta$ is true. Thus, according to $\neg \Pi$, we have

$$\left\{ (t_K^d - t_1^a) P_{\max} \geq \sum_{i=1}^K P_i^{\text{req}} \mid \forall (t_1^d < t_2^a) \wedge \dots \wedge (t_K^d < t_{K-1}^a) \right\}. \quad (21)$$

Combining (16b), (20), and (22), we have

$$\sum_{t=t_1^a}^{t_K^d} P_{ch(m,t)} \leq \sum_{i=1}^K P_i^{\text{req}} \leq (t_K^d - t_1^a) P_{\max}. \quad (22)$$

Therefore, $\exists P_{ch(m,t)}$ to make the constraint (15d) established. Then, since we have $t_K^d < t_{K-1}^a$, the relationship can be obtained by combining constraint (15d)

$$\begin{cases} \sum_{t=t_i^a}^{t_{i+1}^d} P_{ch(m,t)} \leq \sum_{t=t_i^a}^{t_i^d} P_{ch(m,t)} = P_i^{\text{req}} \\ \sum_{t=t_i^d}^{t_{i+1}^a} P_{ch(m,t)} \leq \sum_{t=t_{i+1}^d}^{t_{i+1}^a} P_{ch(m,t)} = P_{i+1}^{\text{req}} \end{cases} \quad \forall i \in K \quad (23)$$

where (23) means EV i is charged less power in $[t_i^a, t_i^d + 1]$ than in $[t_i^a, t_i^d]$ and EV $i + 1$ is charged less power in $[t_i^d, t_i^d + 1]$ than in $[t_i^a + 1, t_i^d + 1]$. Both EV i and EV $i + 1$ are charged at the same charger and EV $i + 1$ is the subsequent of EV i . Therefore, constraint (15a) is satisfied. Obviously, constraint (15c) is easy to establish, so all constraints of the problems (14) and (15) are satisfied, which means $\neg \Theta$ is true when $\neg \Pi$. Therefore, the statement $\Pi \Leftrightarrow \Theta$ is true.

The solving algorithm flowchart of the proposed BP problem is described in Fig. 3. Follow constraint (17) to generate δ as the individual and input it into the lower-level model. Then verify whether the problem (14) and (15) is solvable under each fixed δ following Theorem 1. The solvable problem will take δ as a known parameter and solve the linear problem with decision variables $P_{ch(m,t)}$. The unsolvable situation will return “break” information to the upper level. If there are solvable individuals, the minimum charging cost G of each individual, outputted from lower level, will be their fitness values, and the individuals with smaller G values are more likely to reproduce offspring. Record the individual with the smallest G value at this generation. Then individuals who are

able to produce offspring, generate new population through crossover and mutation, and then repeat the above steps. Until the predetermined generation is reached, stop the calculation and output the best individual as the result. Then output the best chargers assignment matrix δ and its corresponding charging power matrix P_{ch} . If all individuals in the population are unsolvable, output the best solution recorded before. If no best solution has been recorded, then the number of EVs exceeds the serviceability of the charging station.

Algorithm 2 EV Charging Power Calculating Algorithm

1: **Input:** P_{ch} , δ , VEC , M , N , T , P_{max}

2: **Output:** P_{ev}

3: **Initialization:** $P_{ev} = \text{zeros}[N, T]$ ($N \times T$ null matrix), $ICV_m = []$

4: **Procedure** ICV_m calculation

5: **for** every charger $m = 1: M$

6: **for** every EV $i = 1: N$

7: **if** $\delta_{(m,i)} = 1$ **then**

8: $ICV_m = [ICV_m, i]$

9: **end if**

10: **end for**

11: **end for**

12: **return:** ICV_m

13: **end Procedure**

14: **Procedure** P_{ev} calculation

15: **for** every charger $m = 1: M$

16: $n = 1$

17: **for** every time slot $t = 1: T$

18: $j = ICV_{m(n)}$

19: **if** $P_j^{req} > 0$ **then**

20: **if** $P_{rem} > 0$ **then**

21: $P_{ev(j,t)} = P_{rem}$

22: $P_j^{req} = P_j^{req} - P_{rem}$

23: **else**

24: $P_{ev(j,t)} = P_{ch(m,t)}$

25: $P_j^{req} = P_j^{req} - P_{ch(m,t)}$

26: **end if**

27: **else**

28: $P_{ev(j,t)} = P_j^{req}$

29: $P_{rem} = P_{ch(m,t)} - P_j^{req}$

30: $n = n + 1$

31: **end if**

32: **end for**

33: **end for**

34: **return:** P_{ev}

35: **end Procedure**

Combining the optimal charger assignment matrix δ and charger power matrix P_{ch} , the EV power matrix P_{ev} , which shows the specific charging power of each EV at each time slot, can be obtained. The detailed procedure for calculating P_{ev} is given in Algorithm 2. Define ICV_m as the vector that contains index of charging EV at charger m . $ICV_{m(n)}$ represents the n th element in vector ICV_m . According to Remark 4, one charger can only serve the subsequent EV when the previous EV finished charging, so the charging power

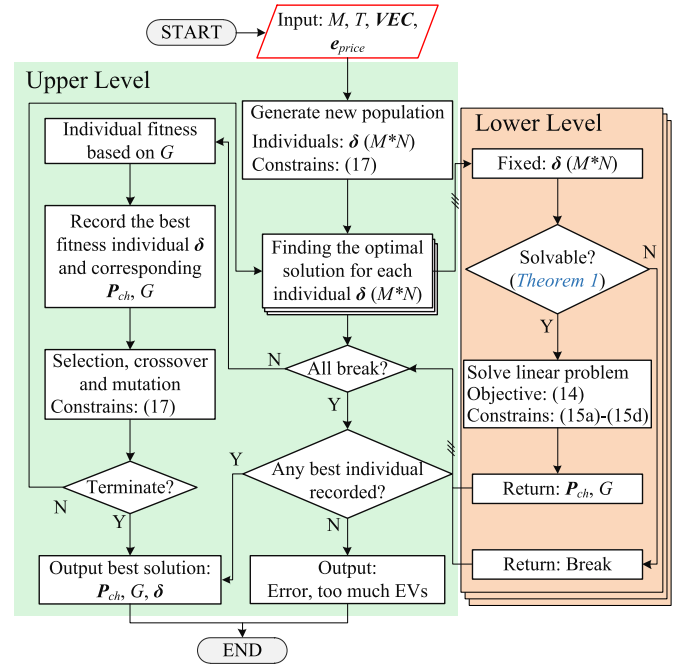


Fig. 3. Algorithm flowchart.

of one charger will be allocated in accordance with charging sequence and required power of EVs. By combining charger assignment matrix δ and EV power matrix P_{ev} , The specific charging scheme, such as each individual EV charging at which charger and the specific charging power at each time slot, can be obtained.

IV. PERFORMANCE EVALUATION

Since the genetic algorithm is adopted in the POCS algorithm, the performance of the proposed algorithm is discussed. As described above, by establishing the BP model, we can greatly reduce the searching range of the genetic algorithm. As shown in Fig. 4, directly using the normal genetic algorithm to find the results leads to searching both δ and P_{ch} at the same time, where P_{ch} belongs to positive natural numbers. That means the calculation time will be long and the results will be far away from the optimal solution. By adopting the solving method proposed in this article, the searching range is reduced to only search δ . The value range of P_{ch} is a real number greater than 0, and the range of δ is only selected between 0 and 1, so the actual possible types of chromosomes is reduced from infinity to a finite number. On this basis, when considering constraint (17), the possible types of chromosomes will be much fewer. Because the searching range has been reduced significantly, the computing speed and accuracy will be increased.

Monte Carlo analysis is adopted to find the average generation for getting the best fitness value and the average calculating time under different number of chargers M and different number of EVs N . Set N to a fixed value 15, calculate the average required generation to obtain the optimal solution and the average calculation time under different M . Similarly, set M to a fixed value 6, calculate the average required generation to obtain the optimal solution and the average

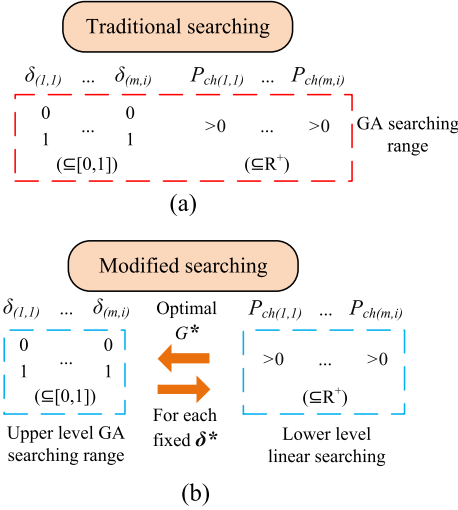


Fig. 4. Searching range of (a) traditional genetic algorithm and (b) modified genetic algorithm.

M/N	Average generation of best fitness value	Average calculating time
4/15	2.5	0.081s
6/15	3.8	0.088s
8/15	4.1	0.102s
10/15	4.5	0.110s

M/N	Average generation of best fitness value	Average calculating time
6/15	3.1	0.084s
6/20	3.9	0.088s
6/25	4.7	0.105s
6/30	4.4	0.095s

Population size: 50, Generations: 20,
Crossover rate: 50%, Mutation rate: 5%

calculation time under different N . The results and parameters of the genetic algorithm are shown in Table I. It can be seen that when the number of chargers increases, the calculation time will also increase. This is caused by the increase in chromosome length (increase in the number of rows of δ). Similarly, when the number of EVs increases, the calculation time will increase. But when the number of EVs exceeds the serving capacity, the calculation time will decrease. This is because according to Theorem 1, we can quickly determine whether the lower-level model has a solution, which can reduce the calculating time. According to the performance test, we can consider that the parameters setting of the genetic algorithm is reasonable, and it can find the optimal solution in an acceptable time.

In order to demonstrate the performance of our POCS algorithm, the UCS, and the most widely adopted optimal charging without chargers assignment scheduling (WCAS)¹ is

¹The WCAS method is the most common scheduling method when considering the charging station capacity limitations. However this method ignores the specific assignment of charger for each individual EV, which means it cannot figure out the chargers assignment matrix δ . Thus, in this article, we call it without chargers assignment scheduling (WCAS). The objective function of WCAS in this article for comparison is (8) in this article and the constraints are based on the model in [28].

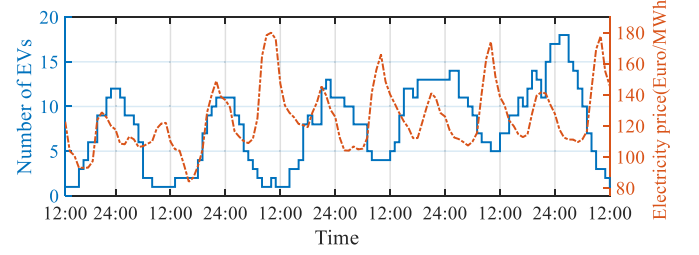


Fig. 5. TOU electricity price and number of EVs dwelling at charging station.

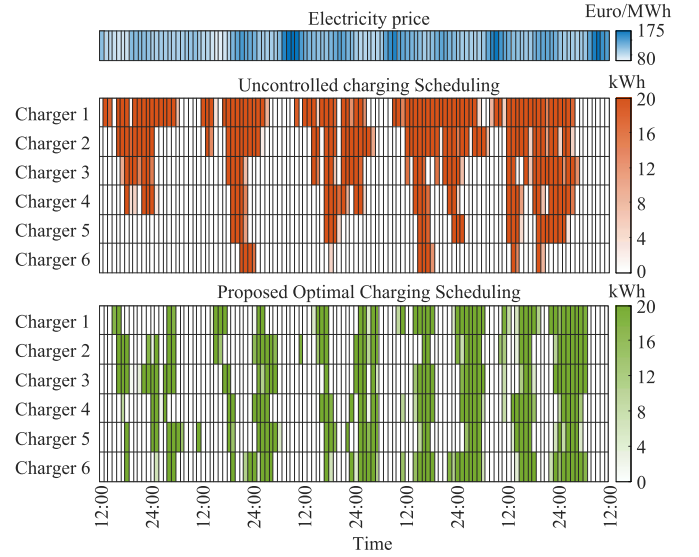


Fig. 6. Heatmap of the energy provided by each charger at different times.

taken for comparisons, one charging station for the logistics EVs is considered as a case study. In addition, to prove the universality of the proposed method, Monte Carlo simulation is adopted to verify the efficiency. The serviceability under different charging method is compared, the impact of the number of EVs on the charging cost under different charging method is analyzed, and the economized charging cost rate of the POCS method is compared with the UCS method.

A. Case Study

The number of chargers is $M = 6$, all EVs and chargers are of the same type and the maximum charging power per hour provided by each charger is $P_{\max} = 20$ kW/h. The battery capacity of the EV is set as $\text{Cap} = 100$ kWh. The optimization time slot t is set as one hour while the total time slot number T in an optimization duration is set to 24 h. Because EVs usually transport cargo during the daytime and return to the logistic center at night, the optimization duration starts from first day 12:00 to the next day 12:00. In total 100 EVs were charged in five days. The time of used electricity price follows the Germany electricity spot prices [33]. Both the number of EVs stay in the charging station at each time slot and the TOU price are shown in Fig. 5.

The heatmap of power provided by each charger at different times is shown in Fig. 6. It can be found that the detailed charging process about how each charger provides energy to

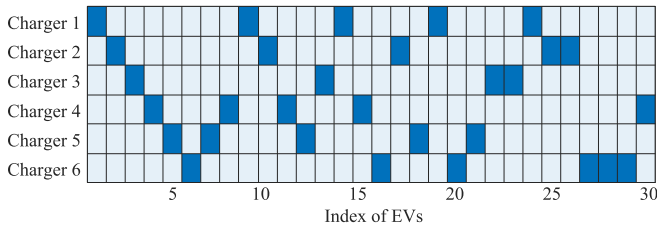


Fig. 7. Selection of chargers in a topical day.

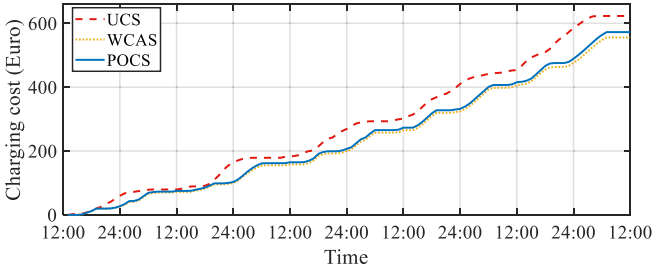


Fig. 8. Cumulative charging cost under different charging methods.

each EV has been established by the proposed algorithm. The charging power under POCS is more dispersed in the heatmap. In the POCS method, the chargers are always avoiding to supply energy when the electricity price was at a peak value. In contrast, by adopting the UCS, EVs start charging as soon as they find the idle charger, thus, the economics of EVs energy supplying in this case is much lower than after scheduling optimization. In addition, the POCS can find out the exact charger index of each specific EV, Fig. 7 shows the charger selection of each EV in a topical day where the deep color means this EV was charging at the specific charger.

Fig. 8 expresses the cumulative charging cost under three different charging methods. It can be found from Fig. 7 that the cost by adopting WCAS is the least because it simulates an ideal scene and ignores the specific charger assignment scheme for each individual EV. With the help of the POCS, the charging cost is reduced significantly compared with the UCS, and the result is very close to the ideal WCAS condition. Meanwhile, a specific chargers assignment matrix δ cannot be given in the ideal WCAS scenario, so we believe that the POCS can not only reduce charging costs but also provide a more realistic charging operation scheme.

B. Serviceability of the Charging Station

In the realistic operation of the charging station, it is necessary to ensure all EVs are fully charged. However, the number of chargers is limited and each EV requires a specific time for charging, so the ability to provide charging services per day of a charging station is limited by the number of chargers. Because the power requirement is stochastic, the serviceability of the charging station is discussed by the Monte Carlo analysis. The optimization duration T is set as 24 h from 12:00 to the next day 12:00. Generate one set of VEC with a certain number of EV N by following normal distribution, as described in Section II. And then, input the generated EVs set VEC to the optimal charging scheduling model and the

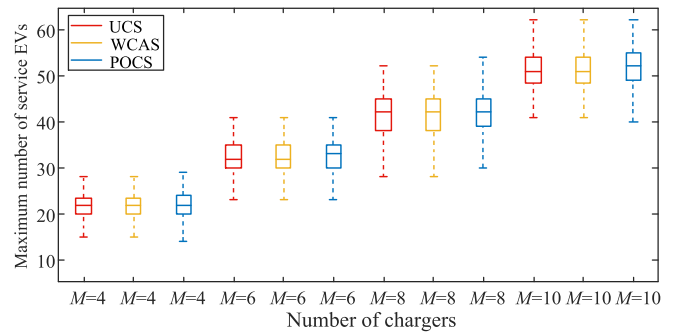


Fig. 9. Distribution of serviceability after Monte Carlo analysis.

uncontrolled charging model. If all EVs are fully charged, increase the number of N , and then generate another set VEC and repeat the above steps continuously. Once not all EVs are fully charged, record current N as the serviceability. By repeating the above process 500 times, the boxplot of the serviceability under different charger number is shown in Fig. 9.

Obviously, as the number of chargers increases, the number of EVs that can be charged per day also increases. In addition, the serviceability under the POCS is very close to the serviceability under the UCS and WCAS, which means the proposed charging scheduling method can guarantee the maximum number of EVs to complete charging.

C. Charging Cost Under Different Charging Methods

The charging cost under two charging methods with the increase of charging EVs is also analyzed by Monte Carlo simulation. For a certain number of N , generate a set of VEC , then input it to the optimized charging scheduling and the uncontrolled charging model to calculate the different charging costs. Repeat this process 500 times, then the charging costs of the current number of N under these two different charging methods are computed.

The simulation results are shown in Fig. 10. The yellow dotted line and green dotted line represent the price range of POCS and UCS, respectively. Meanwhile, the blue line and point mark the mean charging cost under POCS and the red line and point mark the mean charging cost under UCS. The mean cost under WCAS is drawn with a black dotted line. Four scenarios with different number of chargers are simulated. Evidently, regardless of the number of chargers, charging with the POCS can reduce the charging cost significantly than the UCS method. Similar to the result in Fig. 8, the cost under POCS is very close to the result under ideal WCAS condition.

D. Efficiency of the POCS Algorithm

By adopting the POCS, the reduced charging cost comparing with the UCS cost is shown in Fig. 11. It can be found that when the number of EVs is not large, the optimal charging cost is close to the UCS cost, because the number of charging EVs that can be rescheduled is small. In contrast, when the number of EVs arrives at a certain number, the optimization is more obvious. However, when the number of charging EVs continues to increase, the optimized charging cost and uncontrolled

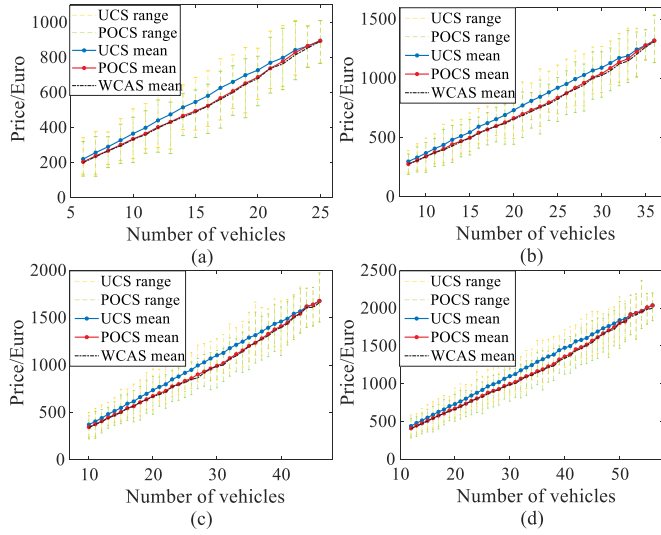


Fig. 10. Charging cost under different charging methods with the increase of charging vehicles. (a) Charger number $M = 4$. (b) Charger number $M = 6$. (c) Charger number $M = 8$. (d) Charger number $M = 10$.

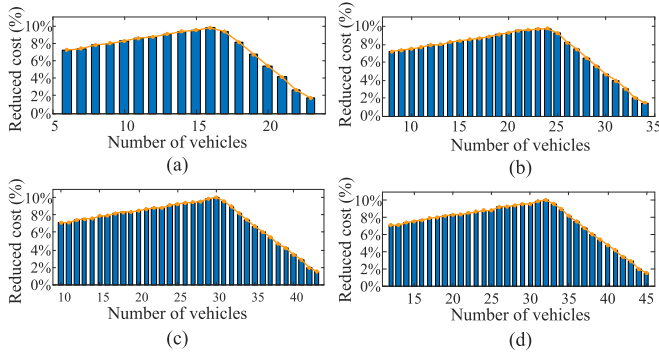


Fig. 11. Reduced charging cost when adopting POCS compared to UCS. (a) Charger number $M = 4$. (b) Charger number $M = 6$. (c) Charger number $M = 8$. (d) Charger number $M = 10$.

charging cost will become closer. This is because the number of EVs is close to the charging station's serviceability, in order to ensure that each EV is fully charged before departure, there is few extra margin for charging scheduling optimization. Through the analysis of the efficiency of the proposed charging scheduling method, the number of chargers in the charging station can be installed according to the actual number of EVs operating every day.

V. CONCLUSION

Considering the limited number of chargers, the charging process of a charging station is modeled. A UCS process is designed to reflect the working way of each charger when no extra optimization method is implemented. Then, an optimal charging scheduling method is proposed by reasonably arranging the charging time and chargers for each EV. Through scheduling the charging power within EVs' available charging time by responding to the TOU electricity price, the proposed method can reduce the charging cost as much as possible while ensuring that all EVs are fully charged. Besides, we analyzed

the serviceability of the charging station with the limited chargers and compared the economized charging cost under different number of charging vehicles and different number of chargers. The POCS method is compared with the UCS method and the WCAS method. Through extensive simulations, it has been shown that with the POCS algorithm, the charging station can not only schedule the charging process more efficiently but also provide a more detailed optimal charging scheme.

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